Geospatial Technology

Louisville, KY

2023

3612-1

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Interpretation and explanation of the challenge

Average global temperatures have raised by an average of 1.2° Celsius since the year 1880. After climatologists and other scientists acknowledged a connection between this rise in temperature and the fall in air quality, the United States government issued the Clean Air Act of 1970. Since this step towards a healthy environment, these temperatures have been projected to reduce within preceding years; however, global temperatures and poor air quality are still a major problem. Around 90% of the heat released by greenhouse gasses is absorbed by the ocean; but that does not mean high temperature and poor air quality are not a problem on land.

A phenomenon known as air stagnation describes the act of air accumulating over a given area and absorbing all of the pollutants from that area. When this happens, temperatures begin to rise, pollutants begin to concentrate, and air quality becomes poorer. Our team's original goal was to analyze how evident this phenomenon of air stagnation was in Tennessee. Tennessee has many changes in elevation, humidity, temperature and population across the state, making it an interesting subject to analyze.

To complete the task of comparing air quality to heat, we first created a map of heat across the state of Tennessee using the program QGIS with no prior experience. We then observed ozone concentration in 21 of the 95 counties in Tennessee. We exported the data from each layer of the map (heat and ozone) at the same overlaying points to an Excel document, where the data were analyzed using regression. The result from this regression plot was not significant enough to adequately draw a conclusion about air stagnation in Tennessee.

Following the failure of this initial analysis, our team decided to instead analyze a value called Air Quality Index (AQI). We analyzed this value from 23 of the 95 counties in Tennessee. The data from this value was placed in QGIS over top of the heat layer in a delimited text layer, just as the ozone was. We exported the data to an Excel document, but this time, instead of exporting a single point from each of the data points, we exported a mean of the temperature in an ~11km diameter circle and compared it to the point data from AQI. A mean was taken rather than a median, because a mean ensures that any outliers in data will still affect the datum as much as possible. We used a regression plot to compare the extracted values. We were able to draw a conclusion from the result of this analysis, though it is still fairly insignificant.

Because of the insignificance of this conclusion, our team decided to compare the result to another factor that could contribute to increase in AQI (population), then determine which factor contributes to a rise in AQI more in Tennessee. In order to do this, we created a second map of Tennessee using population density data, then overlayed the same layer of AQI from 23 of the 95 counties in Tennessee. The population density was extracted in the same ~11km circles centered at the AQI points and analyzed in a regression plot. This analysis was satisfactory and allowed for a complete conclusion.

Following our state conference, our team decided to add an additional factor to the analysis – humidity. This factor was measured in the same way temperature and population were measured in Tennessee. We also added analysis of all three factors in the state of Utah to see how much humidity affects the overall impacts of air stagnation and humidity on poor air quality.

Collected Geospatial Data

Daymet: Monthly Climate Summaries on a 1-km Grid for North America, Version 4 R1. Thornton, M.M., R. Shrestha, Y.Wei, P.E. Thornton, S-C. Cao, and B.E. Wilson. 2022. ORNL DAAC, Oak Ridge Tennessee, USA. <doi.org/10.3334/ORNLDAAC/2131>. Data for temperature downloaded from the Oak Ridge National Laboratory Distributed Active Archive Center for Biogeochemical Dynamics on 6 November 2022. 1km x 1km spatial resolution for the year 2019 in geoTIFF file formats. Data were projected to WGS84 for analysis.

Environmental Protection Agency: Daily Air Quality Summary Statistics. Summary of the Daily Air Quality of Tennessee in the year 2019 regarding Ozone. United States Environmental Protection Agency. <www.epa.gov/outdoor-air-quality-data/download-daily-data>. Downloaded from EPA on 26 November 2022. Comma Separated Values for the year 2019.

Environmental Protection Agency: Daily AQI (Air Quality Index): Daily AQI by TN County 2019. United States EPA. <aqs.epa.gov/aqsweb/airdata/download_files.html#AQI>. Downloaded from EPA on 30 December 2022. Comma Separated Values for the year 2019.

United States Census Bureau: Cartographic Boundary Files. Cartographic Boundary of each of the 50 US States. <www2.census.gov/geo/tiger/GENZ2021/shp/cb_2021_us_state_500k.zip>. Downloaded from US Census Bureau on 14 December 2022. Shapefile for the year 2019.

WorldPop: Population Density: The spatial distribution of population density in 2020 based on country total adjusted to match the corresponding UNPD estimate, United States. University of South Hampton, UK. <data.humdata.org/dataset/worldpop-population-density-for-united-states-of-america>. Downloaded from Humanitarian Data Exchange on 30 December 2022. 1km x 1km spatial resolution for the year 2020 in geoTiff file formats. Tennessee file used.

Environmental Protection Agency: Daily AQI (Air Quality Index): Daily AQI by UT County 2019. United States EPA. <aqs.epa.gov/aqsweb/airdata/download_files.html#AQI >. Downloaded from EPA on 7 April 2023. Comma Separated Values for the year 2019.

WorldPop: Population Density: The spatial distribution of population density in 2020 based on country total adjusted to match the corresponding UNPD estimate, United States. University of South Hampton, UK. <data.humdata.org/dataset/worldpop-population-density-for-united-states-of-america>. Downloaded from Humanitarian Data Exchange on 30 December 2022. 1km x 1km spatial resolution for the year 2020 in geoTiff file formats. Utah file used.

Daymet: Monthly Climate Summaries on a 1-km Grid for North America, Version 4 R1. Thornton, M.M., R. Shrestha, Y.Wei, P.E. Thornton, S-C. Cao, and B.E. Wilson. 2022. ORNL DAAC, Oak Ridge Tennessee, USA. <doi.org/10.3334/ORNLDAAC/2131>. Data for vapor pressure downloaded from the Oak Ridge National Laboratory Distributed Active Archive Center for Biogeochemical Dynamics on 6 November 2022. 1km x 1km spatial resolution for the year 2019 in geoTIFF file formats.

Analysis of Collected Geospatial Data

How much does humidity influence stagnant air and poor air quality? Which factor influences air quality more – population or stagnant air? These questions were addressed through data including AQI in both Utah and Tennessee, population density in Utah and Tennessee, heat in Utah and Tennessee, and vapor pressure in Utah and Tennessee.

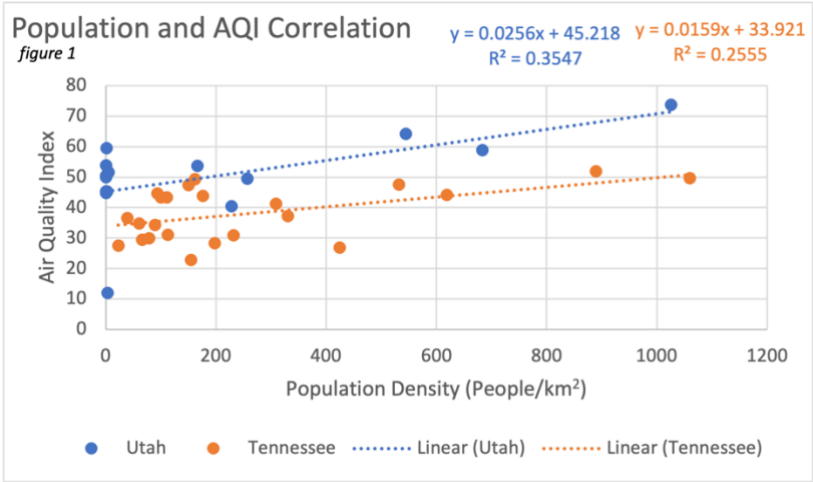


Figure 1 – Population and AQI correlation – consists of two least squares regression lines (LSRLs). An LSRL is a statistical tool that displays a linear relationship between points in a scatter plot – placed by their X (Population Density) and Y (AQI) values. The linear relationship is created by fitting a line to the points that limits the residuals (the vertical distance from the data points to the line itself) as much as possible. The line is called the "least" squares regression line because a line that properly fits a set of data points will minimize the variance in residuals of the chart.

The slope of the orange line states that when the mean population density for a circular area with a diameter of ~22km (0.1 geographical degrees) increases by 1 person per kilometer, it causes an average increase of 0.0159 points on the AQI in Tennessee. The blue equation on the graph describes that when the mean population density for a circular area with a diameter of ~22km increases by 1 person, it causes an average increase of 0.0256 points on the AQI in Utah. The slope of the blue line is steeper than the slope of the orange line, meaning that, if there is enough correlation, population contributes to poor air quality more in Utah than it does in Tennessee.

On every analysis done on this data, the same longitude and latitude values are used. This means that the AQI values for each graph come from the same location and has the same value. The y-intercept of the blue data is 45.218, which is greater than the y-intercept of the orange data, which is 33.921. Air quality is, on average, poorer in Utah than it is in Tennessee.

The value, *R*, is called a correlation coefficient. This value ranges from -1 to 1 and measures the strength of the linear relationship, or the degree of association between the two variables of the LSRL. When the correlation coefficient is 1 or -1, the line displays a perfect positive or negative correlation (Y = X or Y = -X). When the correlation coefficient is 0, there is no correlation. The farther the correlation coefficient is from 0, the better fit the LSRL, and the greater the correlation. The correlation coefficient of the graph representing population and AQI correlation in Tennessee has a value of 0.5055. This means there is a moderate correlation between the two variables of heat and AQI in Tennessee. The correlation coefficient for population and AQI in Utah is 0.5956.

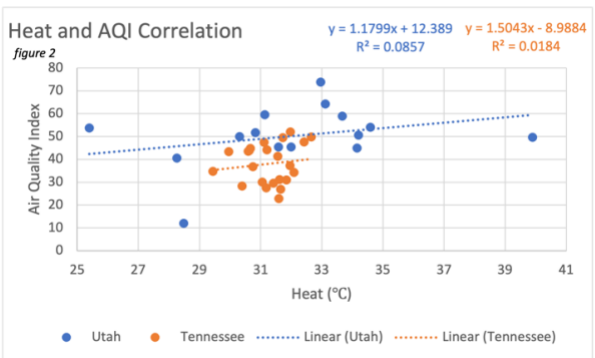
This means there is also a moderate correlation between the two variables of population and AQI in Utah, however, there is slightly more correlation between these two variables in Utah than Tennessee.

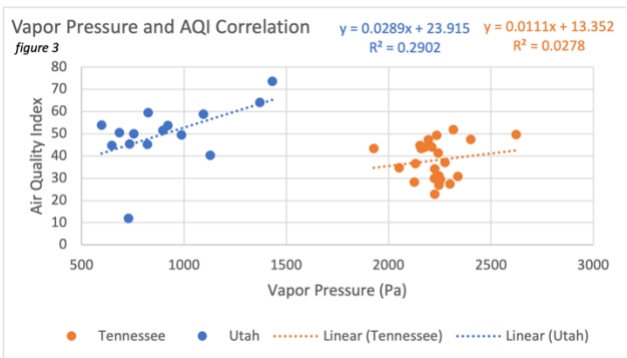
The R^2 value is known as the coefficient of determination, and it is always positive, ranging from 0 to 1 (0% to 100%). The coefficient of determination is a statistical measure explaining the proportion of variance for a dependent variable (AQI) explained through an independent variable (population) in an LSRL. The correlation coefficient shows the strength of a relationship, while a coefficient of determination explains the extent to which an independent variable accounts for in the variance of a dependent variable. In the context of the population and AQI correlation graph in Tennessee, the coefficient of determination tells us that 25.55% of the variation in AQI can be accounted for in the LSRL that matches AQI to heat. In other words, when air quality changes in Tennessee, 25.55% of that change is influenced by population. In contrast, 35.47% of the variance in AQI can be accounted for in the LSRL that matches AQI to population in Utah, or, 35.47% of the change in air quality in Utah is due to population.

An additional test can be done to compare the correlation of the data in the two states. This test is called a confidence interval. A confidence interval conveys how precise a given measurement is. The interval states that within a range between two numbers, the predicted value of a sample of data will fall within that range during a specific percentage of samples. For example, if a 95% confidence interval were taken on the mean of a population, then if many different samples were taken from that population, 95% of the means from the samples would fall within the interval determined.

In context of this data, a confidence interval was taken on the slope of the LSRLs. When a confidence interval is taken for the slope of the orange line that matches population to AQI in Tennessee, it can be stated with 95% confidence that the interval from 0.00358609 to 0.028225 captures the true slope of the LSRL that matches population to AQI. This is considered statistically significant. The lowest value of this interval is not zero (or negative) because it would indicate a possibility of no correlation (or negative correlation) between population and AQI. A 95% confidence interval with two positive boundaries almost guarantees the slope will be positive. Since 0 – meaning no difference – is not in this confidence interval, it can be stated with 95% confidence that an increase in population density is associated with an increase in AQI. This prompts a positive correlation between the two variables of population and AQI in Tennessee.

When a confidence interval is taken for the blue line that matches population to AQI in Utah, it can be stated with 95% confidence that the interval from 0.00490767 to 0.462807 captures the true slope of the LSRL that displays the correlation between population and AQI. This is statistically significant. The lowest value of the interval is not zero, so this 95% confidence interval says that it is very likely that the slope will be positive, as 95% of the time a sample of data correlating temperature and AQI in Utah will result in a positive slope.





The orange data in figure 2 represents the correlation of heat and AQI across the state of Tennessee. The orange equation displayed on the graph describes that if there is any correlation, when the temperature of a circular area with an ~11km diameter (0.05 geographical degrees) increases by 1 degree, there is an average increase of 1.1799-points in the mean AQI. The blue data on the graph represents the correlation of heat and AQI across the state of Utah. The blue equation displayed on the graph describes that, if there is any correlation, when the temperature of a circular area with an ~11km diameter increases by 1 degree, there is an average 1.5043-point increase in the mean AQI. Each slope is positive, meaning that, as long as there is correlation, a higher temperature is associated with poorer air quality.

The correlation coefficient for heat and AQI in Tennessee is 0.136. This means there is a weak – almost no – correlation between the two variables of heat and AQI in Tennessee.

The data in figure 3 describes the correlation between vapor pressure and AQI in both Tennessee and Utah. The orange equation for the graph on the right states that if there is any correlation, for every 1 pascal increase in vapor pressure in a circular area with an ~11km (0.05 geographical degrees) diameter, there is a 0.0111-point increase in the mean AQI. The blue equation for the graph on the right states that if there is any correlation, for every 1 pascal increase in vapor pressure, there is a 0.0289-point increase in the mean AQI for a circular area with an ~11km (0.05 geographical degrees) diameter. If there is correlation between vapor pressure and AQI for Utah and Tennessee, there is a positive association between the two factors.

The correlation coefficient for vapor pressure and AQI in Tennessee is 0.168. This means there is a very weak correlation between the two variables of vapor pressure and AQI in Tennessee. The correlation coefficient for vapor pressure and AQI in Utah is 0.537. This

The correlation coefficient for heat and AQI in Utah is 0.293. This means there is a moderately weak correlation between the two variables of heat and AQI in Utah, however, there is more correlation between these two variables in Utah than Tennessee.

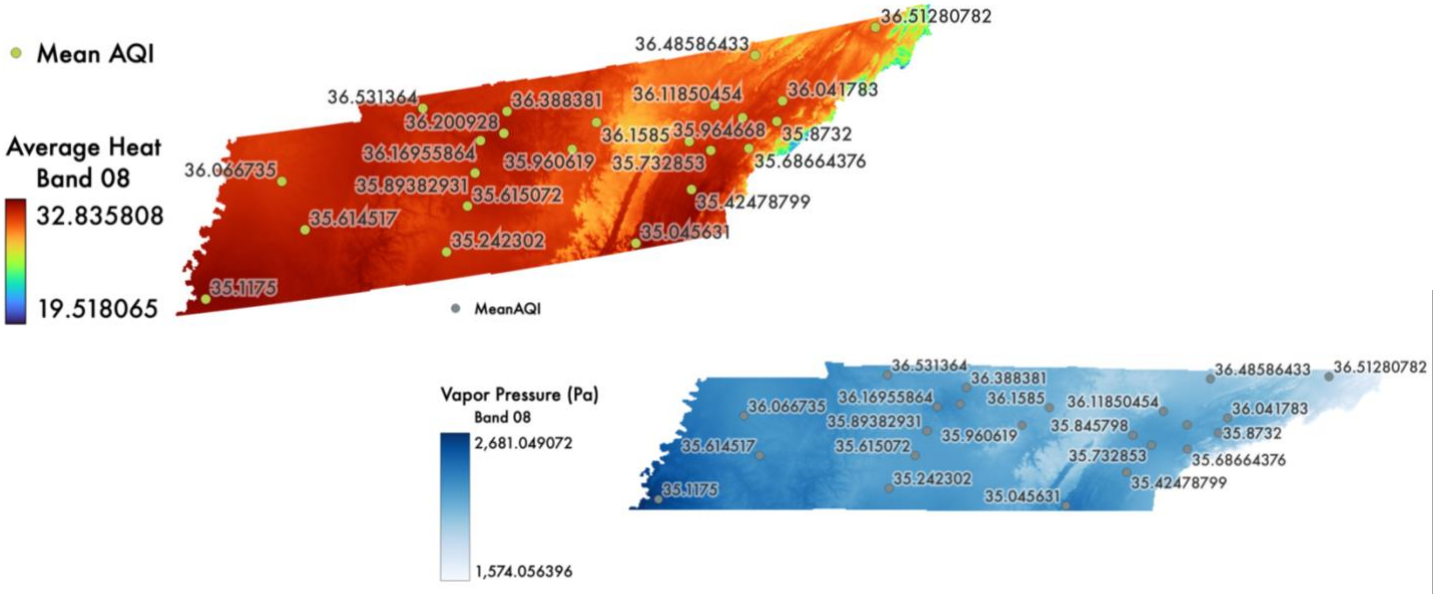
The coefficient of determination tells us that 1.84% of the variation in AQI can be accounted for in the LSRL that matches AQI to heat in Tennessee. In other words, when air quality changes in Tennessee, 1.84% of that change is influenced by heat. In contrast, 8.57% of the variance in AQI can be accounted for in the LSRL that matches AQI to heat in Utah, or, 8.57% of the change in air quality in Utah is due to heat.

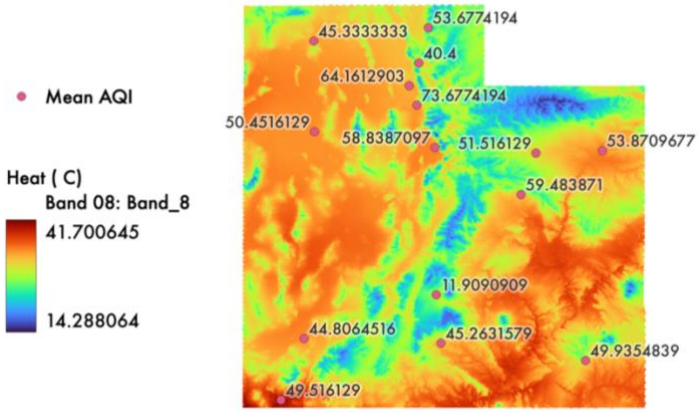
Another important factor to take into account is the range, minimum, and maximum of the data. The range of the orange data is 3.215, or from 29.456 to 32.651. The range of the blue data is 14.499, or from 25.393 to 39.892. The range of the Tennessee heat data is much smaller than the Utah heat data. While Utah can be much colder than Tennessee, it can also be much warmer than Tennessee, allowing for a better understanding of how higher or lower temperatures in Utah affect air quality.

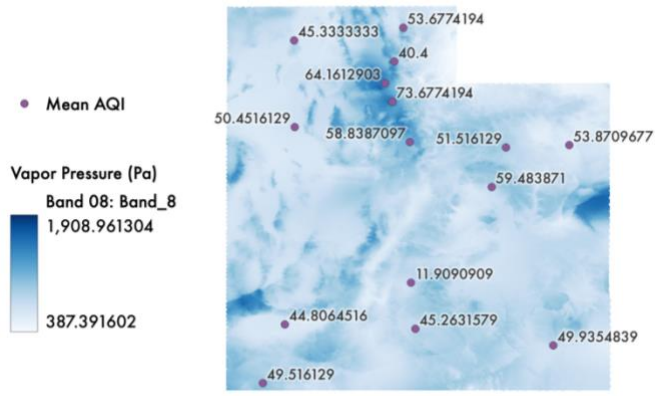
means there is a moderate correlation between the two variables of vapor pressure and AQI in Utah. There is significantly more correlation between vapor pressure and AQI in Utah than there is in Tennessee.

The coefficient of determination for the orange line tells us that 2.78% of the variance in AQI can be accounted for in the LSRL that matches AQI to vapor pressure in Tennessee. In other words, when there is a change in vapor pressure, 2.78% of that change is due to vapor pressure. The coefficient of determination, or R^2, for the blue line tells us that 29.02% of the variance in AQI in Utah can be accounted for in the LSRL that matches AQI to vapor pressure, or, when AQI changes in Utah, 29.02% of that change is due to vapor pressure.

The range, minimum, and maximum of the orange data in figure 3 is 696.356, or from 1,927.569 to 2,623.924. the range of the blue data is 833.050, or from 597.928 to 1430.978. The minimum and maximum values are of more concern than the range of the data itself in figure 3. The air in Utah, on average, has a much lower vapor pressure than Tennessee does. This means that, on average, Tennessee is far more humid than Utah is. This difference allows for understanding of how much vapor pressure/humidity influences stagnant air and poor air quality.







Due to the slopes, coefficients of determination, correlation coefficients, and confidence intervals, a conclusion can be drawn explaining that there is not enough of a correlation between the heat/vapor pressure and air quality to assume that air stagnation is a significant contributor to poor air quality in Tennessee; however, heat/vapor pressure are significant contributors to poor air quality in Utah. 37.59% of the variation in air quality in Utah is due to stagnant air, as opposed to only 4.62% of the variation in air quality in Tennessee resulting from stagnant air. Stagnant air is a larger contributing factor to poor air quality than population in Utah (where only 35.47% of the variance in air quality is due to population), and population is a larger contributing factor to poor air quality than stagnant air in Tennessee (where 25.55% of the variance in air quality is due to population).

The correlations of the population data is significant enough to conclude that population is a significant contributor to poor air quality in many different types of climates. A larger population is indicative of larger urbanized industries and technology, such as factories and cars. These factors produce an abundance of pollutants, including particulate matter, ozone, and carbon monoxide. So, while stagnant air can contribute to poor air quality, population and the factors that come with a large population contribute to poor air quality much more.

~70% of the variation in air quality in Tennessee, and ~27% of the variation in air quality in Utah, is accounted for by factors other than variation from heat, vapor pressure and population as described in this analysis. These variables that are not accounted for in the given study are called confounding variables. Confounding variables in this study may include windiness, lengthening growing season, etc. There are confounding variables in Tennessee are more abundant than they are in Utah. ~30% of these factors in Tennessee, and ~73% of these factors in Utah, however, are population, vapor pressure, and heat.

This analysis concludes that air stagnation is not significantly evident in Tennessee, however, could always have more influence on air quality in other parts of the world, such as Utah. Utah's air quality is significantly influenced by both stagnant air and population, but Tennessee's air quality is influenced primarily by confounding variables (those other than stagnant air and population). The influence of stagnant air on air quality is very limited in Tennessee. The climate and weather patterns of a humid environment analyzed in this study (vapor pressure and heat), such as that of Tennessee, does not determine how poor the air quality is.

Reflection of Project Journal

I. Description of Activities and Timeline of work

9.30.22	Our team begins researching weather phenomena to focus our research on, and how they affect potential areas of focus.
10.12.22	Our team settles on air stagnation for the weather phenomenon of focus, and we begin figuring out what kind of data we will look for, what our map will look like, and what time frame these items should be completed by.
10.29.22	We begin the long process of looking for data, with help from Michele Thornton. Sites like the United States Environmental Protection Agency and the Distributed Active Archive Center for Biogeochemical Dynamics by Oak Ridge National Laboratory.
11.06.22	As we continue searching for data, we find a dataset distributed by ORNL to use for heat analysis in Tennessee.
11.17.22	We continue the search for data.
11.26.22	Using AirNow by the United States Environmental Protection Agency, we find a dataset containing ozone concentration for 21 of 95 counties in Tennessee.
11.27.22	Our team begins using the program QGIS to develop our geospatial map. We import the heat data as a raster layer geoTIFF file.
12.01.22	We import the ozone concentration data as a delimited text layer csv file in our QGIS map. There was no value displayed on the points originally, so we begin figuring out how to display the correct concentration value.
12.09.22	We continue figuring out how to display values on a delimited text layer. A value appeared, but only described the amount of datum under each date, not the maximum amount of ozone concentrated in each area under each date.
12.14.22	After writing a short program within the QGIS software, we are able to manipulate the delimited text layer to display the maximum ozone concentration value within each location. We also use a shapefile to cut out a portion of the raster layer to be shaped like Tennessee.
12.16.22	Using a QGIS plugin, we export an excel file from our map that displays a matched ozone concentration value (the one displayed at each point) to the temperature value located at the same point from the temperature layer.
12.19.22	With the help of Tom Froning, we analyze the exported data in a linear regression plot. The outcome of this data was too insignificant to derive a proper conclusion from it, so we begin thinking of different ways to analyze the same concept of air stagnation.
12.28.22	Our team decides to no longer use ozone concentration in our map, but instead use AQI (air quality index). Additionally, we decide to compare the results from the correlation explaining air stagnation to a correlation between air quality and population.
12.30.22	We find our AQI data from the same source that holds our ozone concentration data. We begin searching for a population dataset, and eventually find one

	describing population density in the state of Tennessee from the United States Census Bureau.
1.04.23	After manually determining the maximum AQI value by each of the 23 locations in the data, we replace the original delimited text layer containing ozone concentration data with the new one containing AQI data. Creating a second QGIS file, we import the population data as a raster layer, and overlay the same delimited text layer containing AQI values.
1.08.23	Using a zonal statistics layer in the air stagnation file, our team creates a layer containing circles centered at the AQI data points with an 11km diameter. From the area of these circles, we take an average value from the heat layer beneath them. We then export this value, along with its matching AQI value to an excel file.
1.09.23	We use a zonal statistics layer on the population and AQI map. Our team exports the data to an excel file.
1.17.23	In the air stagnation file in excel, we create a linear regression plot. While the correlation of this data still is not very strong, the method used to obtain it is statistically sound, and we are able to draw a proper conclusion from it.
1.20.23	In the population and AQI correlation file in excel, we create a linear regression plot. The correlation of this graph was significantly larger than the air stagnation graph, allowing room for a sounder conclusion to be drawn from the data.
1.24.23	Our team writes a conclusion from each of the graphs about the R and R^2 values. After presenting this conclusion to Tom Froning, he introduces a second conclusion we can draw from the data using a confidence interval.
1.25.23	We perform a confidence interval on the slopes of each of the least squares regression lines. We begin writing a second conclusion using this information.
1.26.23	Our team continues to develop conclusions from our data.
1.29.23	We complete writing conclusions from our data.
2.01.23	Our team verbally develops a more refined prediction from our original hypothesis. We begin developing these ideas into more cohesive paragraphs.
2.02.23	We begin writing our final prediction from our original hypothesis.
2.09.23	Our team completes writing our prediction.
2.10.23	Our team properly cites our collected geospatial data.
2.11.23	We properly cite each of our resources.
2.14.23	We begin to write our interpretation and explanation of the challenge.
2.16.23	We write the description of the location factors for our project.
2.17.23	Our team completes writing the interpretation and explanation of the challenge.
2.20.23	We revise all parts of the documentation portfolio. After lots of revision and edits, we are able to confidently say the portfolio is complete.
2.21.23	Our team collects all of the information we will need in order to make our visual infographic map.
2.23.23	We develop the visual aspects of our visual infographic map: color scheme, layout, fonts, paper, etc.
2.26.23	We begin writing the content for our visual infographic map, borrowing a lot of it from the information we already wrote in the documentation portfolio.
2.28.23	Our team prints out all of the information and graphs and maps need for our visual infographic map.

3.04.23	After double checking each of the pieces we printed out for spelling errors, logical fallacies, etc., our team physically puts together our visual infographic map.
3.05.23	We "judge" our visual infographic map based off of the judging rubric. Some of the information we had written down was not completely adequate to the judging standards, so we revised the information.
3.09.23	We "judge" our documentation portfolio. We alter some information and formatting and determine our portfolio as being competition ready.
4.05.23	Following state conference, we review our project and determine what the next steps will be to improve our project as a whole.
4.06.23	We decide to add an additional factor for analysis – humidity. In addition to adding humidity to our analysis, we decide to analyze the three factors (humidity, heat, and population) in Utah as well as Tennessee. We write a list of tasks that must be completed prior to submitting our portfolio for national conference.
4.07.23	With guidance from Michele Thornton, we determine the best dataset to use to find a correlation between humidity and air quality, which measured the vapor pressure in Utah and Tennessee. We also get the data for heat and population in Utah.
4.08.23	We begin researching what vapor pressure is, and how it can affect stagnant air and air quality.
4.09.23	We complete the heat and AQI map for Utah.
4.10.23	We complete the population and AQI map for Utah.
4.13.23	We begin the vapor pressure and AQI map for Tennessee.
4.14.23	We complete the vapor pressure and AQI map for Tennessee and begin the vapor pressure and AQI map for Utah.
4.16.23	We complete the vapor pressure and AQI map for Utah.
4.18.23	Using a buffer layer and zonal statistics tools in QGIS, we export the data correlating heat and AQI in Utah to Excel, as well as the data correlating population and AQI in Utah to Excel.
4.20.23	Using a buffer layer and zonal statistics tools in QGIS, we export the data correlating vapor pressure and AQI in both Utah and Tennessee to Excel.
4.23.23	In Excel, we use the data we exported to create three linear regression charts.
4.25.23	We begin writing and revising the Analysis of Collected Geospatial Data.
4.26.23	We continue writing and revising the Analysis of Collected Geospatial Data.
4.27.23	We finish writing and revising the Analysis of Collected Geospatial Data and move on to writing and revising the Written Explanation of Prediction.
4.30.23	After completing the Written Explanation of Prediction, we begin writing and revising various parts of the portfolio.
5.02.23	We continue writing and revising various parts of our portfolio.
5.06.23	We continue writing and revising various parts of our portfolio.
5.08.23	We finish writing all parts in our portfolio and began revising and editing each section.
5.10.23	Using the judging rubric, we "judge" our portfolio and revise it.
5.13.23	We begin creating our new Visual Infographic Map.
5.14.23	We continue creating our new Visual Infographic Map.
5.15.23	We complete our Visual Infographic Map.

5.16.23	Using the judging rubric, we "judge" our Visual Infographic Map.
5.18.23	After some final readthroughs and revisions, we complete our portfolio and Visual Infographic Map.

II. Description of Location Factors

Our area of focus for this project is both Tennessee and Utah. In making this decision, we considered only the Continental United States (CONUS). We broke down this very large subject area into its smaller regional sections. We first eliminated the Midwest and Northwest as maps of air stagnation show that the phenomenon is not as evident in these regions as the other regions of the United States. The Northeast, we determined, was too cold to analyze an impact of its warm temperatures. Humidity is more evident in the Southeast than in the Southwest where heat is more "dry." Both regions, however, have an abundance of heat. Analyzing both regions would allow us to see the impact of air stagnation on similarly heated yet contrastingly humidified areas. In the Southeast, we chose Tennessee because of the difference in elevation from one side to the other because of the Great Smokey Mountains in East Tennessee. In the Southwest, we chose Utah because of its differing elevations, temperatures, and populations across the state.

Luckily, we were able to find plenty of data for both of these locations. Tennessee does not show very evident air stagnation; however, Utah does. If we were to expand our investigation even further, we could possibly determine where in the entire CONUS is air stagnation most evident, why it is more evident there rather than other locations, how it affects other weather factors and phenomena in the United States, and how much population influences air quality in comparison to heat.

Written Explanation of Prediction

The sun projects heat onto the Earth. This heat is absorbed by the earth's surface, and the warm air resides just below the ozone layer. However, because of density, hot air rises. While hot air is rising and cool air is sinking, the air circulates, and wind picks up. If temperatures are warm enough, air becomes stagnant. Air stagnation occurs as elevated temperatures trap air (including its pollutants) in the lower atmosphere. As temperatures rise, air quality exacerbates, causing the atmosphere to deteriorate and health problems for those residing in areas affected by stagnated air.

As temperatures increase, air molecules absorb heat. In humid areas, the molecules absorb water. In polluted areas, the molecules absorb pollutants in the air. Eventually, air molecules reach their carrying capacity, and can no longer absorb any more matter. In areas with more humidity, air molecules must commit more carrying capacity to water vapor and heat. This would mean that there is less room in a given air molecule for pollutants. From this information, humidity should lead to better air quality; however, there are a few factors this information is not considering. While air molecules can hold liquid water in them, water can be found in the air in its gaseous phase – water vapor. As temperatures rise, water begins converting from its liquid phase to its gaseous phase, "freeing up space" in a given air molecule for pollutants. Water molecules in the air can also carry pollutants in them.

When wind speeds are high, air molecules and airborne water molecules move together in the wind. Air with less water concentration moves into the area, and evaporation rate increases to compensate for the lack of water vapor in the air. The water that evaporates during high wind speeds quickly moves on from the area it originated, creating a constant cycle of high rates of evaporation. When wind speeds are low, however, evaporation rates slow down. Water is still evaporating, but because the air is not moving as much, the concentration of water in the air continues to increase.

Vapor pressure is the total amount of the pressure in the atmosphere due to water vapor. The more water vapor in the air, the higher the vapor pressure will be. So, an increase in vapor pressure is indicative of stagnant air, which leads poor air quality. There should be a positive, direct, linear relationship between the factors of vapor pressure and poor air quality.

While many different pollutants contribute to poor air quality, two of the most prominent include ozone (O_3) and particulate matter ($PM_{2.5}$ and PM_{10}). Ozone is found in both the stratosphere and troposphere. When ozone is found in the stratosphere, it is beneficial to the environment on Earth. It reduces the amount of harmful ultraviolet radiation from reaching Earth's surface. When ozone is found in the troposphere, or at the "ground level," it is formed through a reaction of two chemicals (volatile organic compounds and nitrogen oxides) that typically requires heat and sunlight to take place. Ozone is a large contributor to smog, which occurs most frequently in summer months; however, in southern mountainous regions – such as Tennessee – smog occurs in colder months as well. Almost all the ozone found at the ground level is result of manufactured volatile organic compounds and nitrogen oxides, derived from gasoline, automobile use, power plants, turbines, and other industrial sources.

Particulate matter is made of fine particles ($PM_{2.5}$ is defined as a particle less than 2.5 microns thick) that result primarily from vehicle exhaust and burning fossil fuels. Particulate matter contributes to smog just as ozone does. Regarding data used in this project, the analysis used a combination of the two of these pollutants, as well as carbon monoxide, sulfur dioxide, and nitrogen dioxide.

A value, called the air quality index (AQI), developed by the Environmental Protection Agency, falls in a range from 0 to 500. The higher the value, the worse quality the air is. Any value lower than 50 indicates good air quality. The AQI value is indicative of a health-based national ambient air quality standard, monitored by the Environmental Protection Agency. The AQI assigned to each pollutant concentration value has changed since the AQI was first initiated in 1999 – not only because the quality of air has changed, but also because the way more extreme air qualities affect the human body has changed.

Stagnant air should prompt a direct correlation between elevated temperatures and poor air quality. Any data containing temperature and AQI should display a positive correlation between the two variables (as AQI increases, temperature also increases). This is what our team predicted prior to any data collection or analysis.

Air quality can be influenced by hundreds of different factors. In addition to stagnant air, our team analyzed how population affects air quality. As the population density of a given area increases, as does its economic development, urbanization, energy consumption, and transportation. Many of these factors produce pollutants. As more humans exist on earth, more healthy resources are consumed by those people, and more unhealthy waste products – such as pollutants – are exerted. The production of healthy resources takes a toll on the environment. This production requires exceptionally substantial amounts of energy that creates excess waste and pollution. Additionally, humans use resources and machines that are detrimental to the environment, such as simple plastics and other minor undecomposable items, as well as vehicles, power plants, factories, etc.

Just as air stagnation should prompt a direct, positive, linear correlation between heat and AQI and vapor pressure and AQI, population and AQI should also have a direct, positive, linear correlation. The maps and analysis used in this project ask the questions: How much does humidity influence stagnant air and poor air quality? Which factor influences air quality more–population or stagnant air?

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